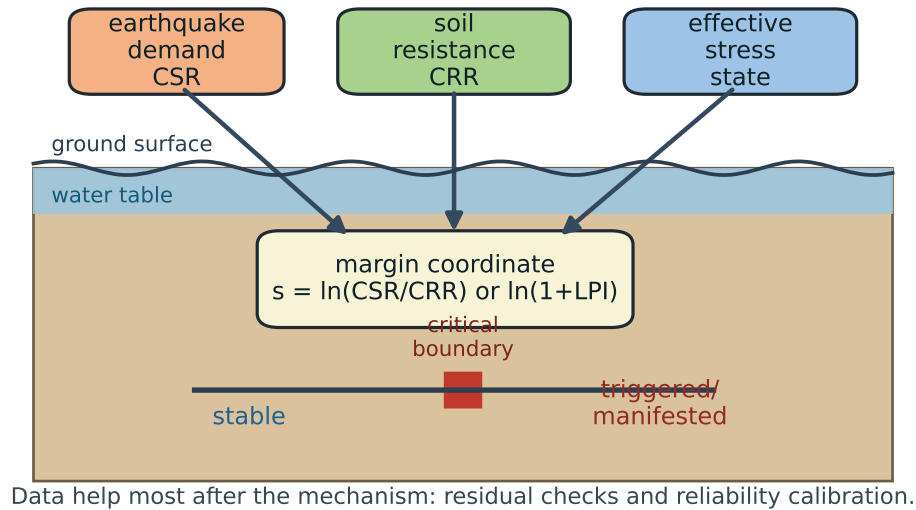
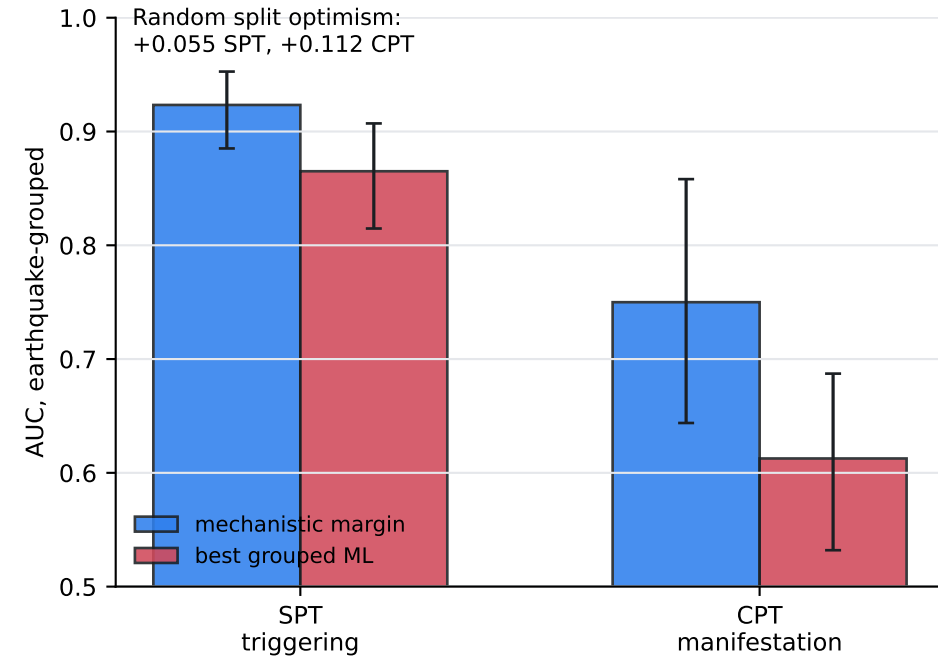


Mechanistic sufficiency and reliability in seismic liquefaction prediction

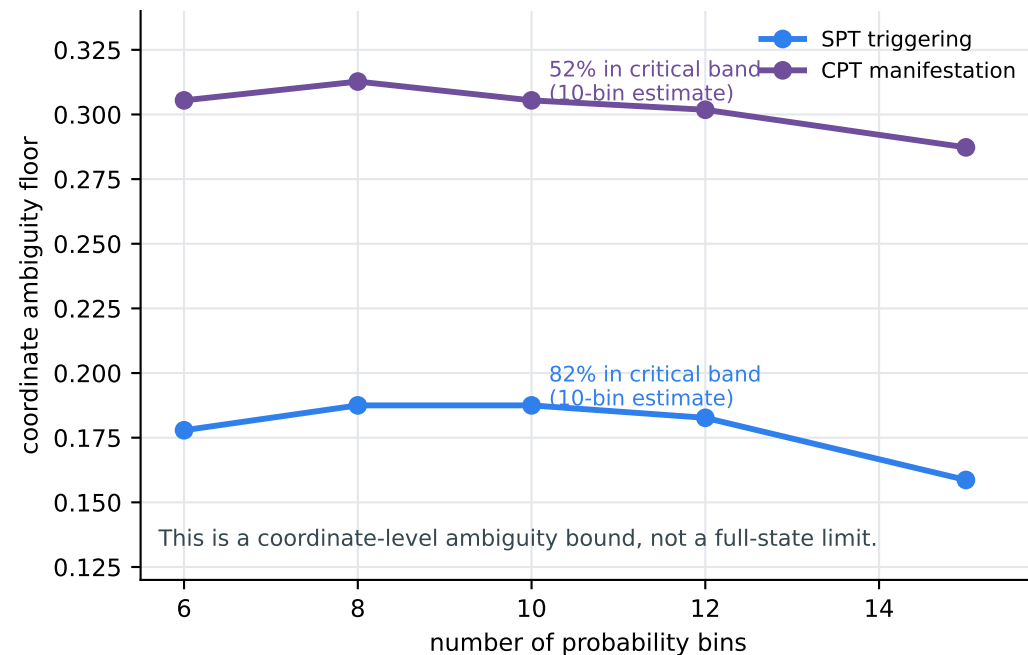
A Effective-stress mechanism as a compressive coordinate



B Grouped validation: mechanism remains the strongest baseline



C Coordinate ambiguity is stable and localized near the boundary



D Mechanism-conditioned conformal moves coverage toward target

